0. Abstraction and Research Process (raw cognition):

Published formulation:

Hand reconstruction → decision architecture → critical interpretation

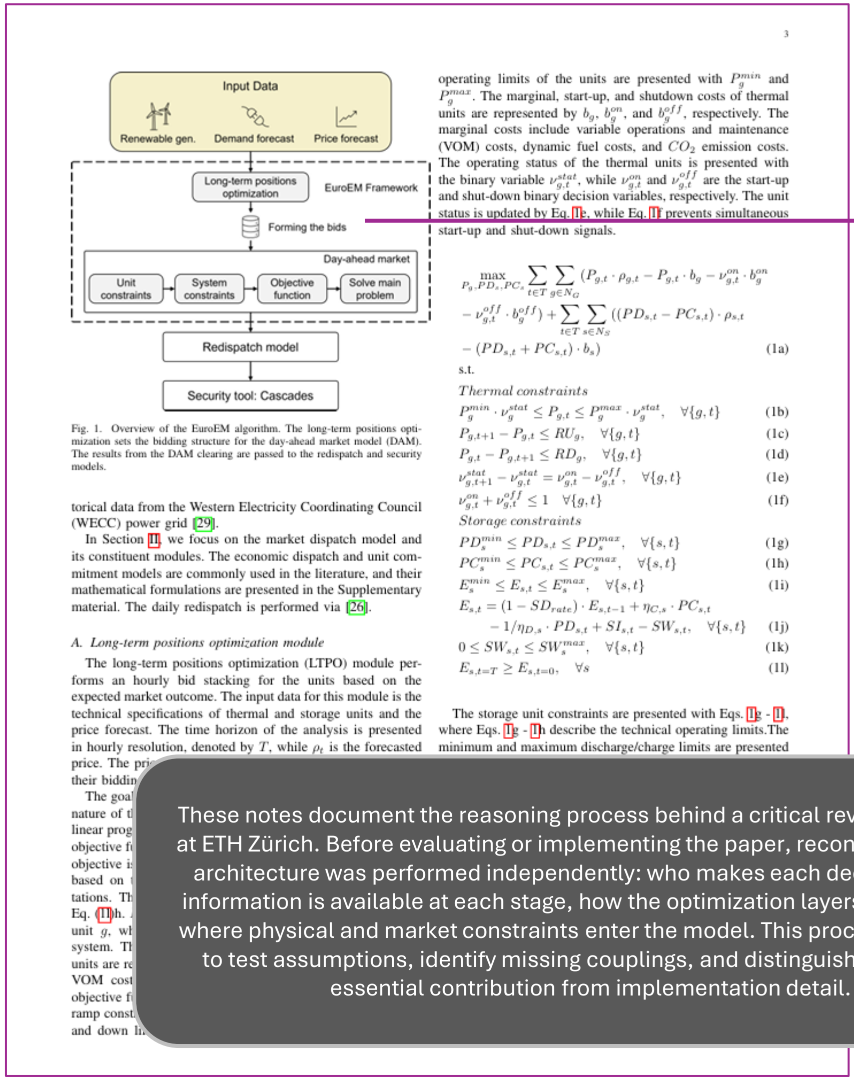

Fig. 1. Overview of the EuroEM algorithm. The long-term positions optimization sets the bidding structure for the day-ahead market model (DAM). The results from the DAM clearing are passed to the redispatch and security models.

torical data from the Western Electricity Coordinating Council (WECC) power grid [29].

In Section II, we focus on the market dispatch model and its constituent modules. The economic dispatch and unit commitment models are commonly used in the literature, and their mathematical formulations are presented in the Supplementary material. The daily redispatch is performed via [26].

A. Long-term positions optimization module

The long-term positions optimization (LTPO) module performs an hourly bid stacking for the units based on the expected market outcome. The input data for this module is the technical specifications of thermal and storage units and the price forecast. The time horizon of the analysis is presented in hourly resolution, denoted by T, while ρ_t is the forecasted price. The pri

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operating limits of the units are presented with P_g^{min} and P_g^{max}. The marginal, start-up, and shutdown costs of thermal units are represented by b_g, b_g^{on}, and b_g^{off}, respectively. The marginal costs include variable operations and maintenance (VOM) costs, dynamic fuel costs, and CO_2 emission costs. The operating status of the thermal units is presented with the binary variable $\nu_{g,t}^{stat}$, while $\nu_{g,t}^{on}$ and $\nu_{g,t}^{off}$ are the start-up and shut-down binary decision variables, respectively. The unit status is updated by Eq. 1e, while Eq. 1f prevents simultaneous start-up and shut-down signals.

$$\max_{P_g, PD_s, PC_s} \sum_{t \in T} \sum_{g \in N_G} (P_{g,t} \cdot \rho_{g,t} - P_{g,t} \cdot b_g - \nu_{g,t}^{on} \cdot b_g^{on} - \nu_{g,t}^{off} \cdot b_g^{off}) + \sum_{t \in T} \sum_{s \in N_S} ((PD_{s,t} - PC_{s,t}) \cdot \rho_{s,t} - (PD_{s,t} + PC_{s,t}) \cdot b_s) \tag{1a}$$

s.t.

Thermal constraints

$$P_g^{min} \cdot \nu_g^{stat} \leq P_{g,t} \leq P_g^{max} \cdot \nu_g^{stat}, \quad \forall\{g,t\} \tag{1b}$$

$$P_{g,t+1} - P_{g,t} \leq RU_g, \quad \forall\{g,t\} \tag{1c}$$

$$P_{g,t} - P_{g,t+1} \leq RD_g, \quad \forall\{g,t\} \tag{1d}$$

$$\nu_{g,t+1}^{stat} - \nu_{g,t}^{stat} = \nu_{g,t}^{on} - \nu_{g,t}^{off}, \quad \forall\{g,t\} \tag{1e}$$

$$\nu_{g,t}^{on} + \nu_{g,t}^{off} \leq 1 \quad \forall\{g,t\} \tag{1f}$$

Storage constraints

$$PD_s^{min} \leq PD_{s,t} \leq PD_s^{max}, \quad \forall\{s,t\} \tag{1g}$$

$$PC_s^{min} \leq PC_{s,t} \leq PC_s^{max}, \quad \forall\{s,t\} \tag{1h}$$

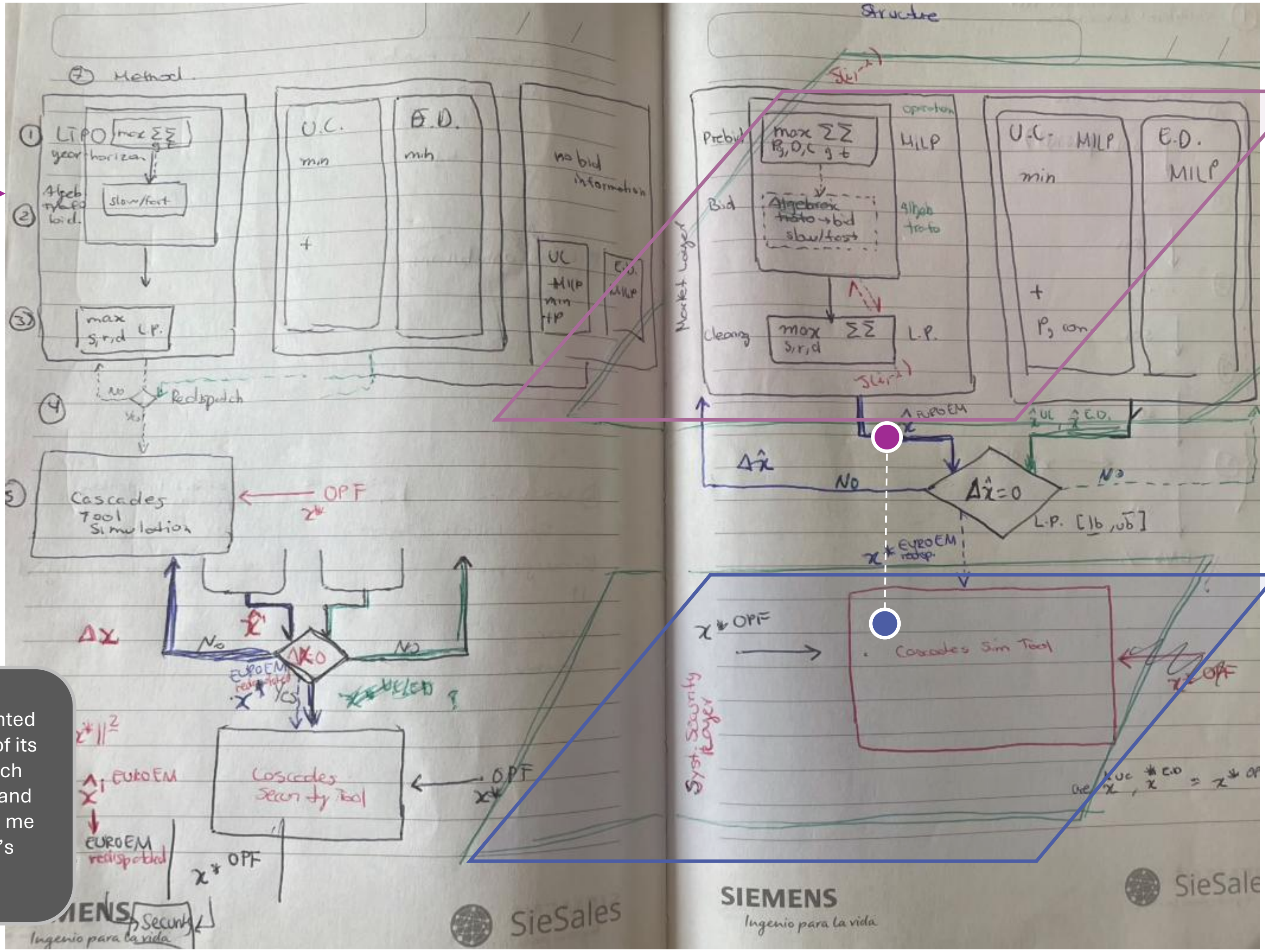
$$E_s^{min} \leq E_{s,t} \leq E_s^{max}, \quad \forall\{s,t\} \tag{1i}$$

$$E_{s,t} = (1 - SD_{rate}) \cdot E_{s,t-1} + \eta_{C,s} \cdot PC_{s,t} - 1/\eta_{D,s} \cdot PD_{s,t} + SI_{s,t} - SW_{s,t}, \quad \forall\{s,t\} \tag{1j}$$

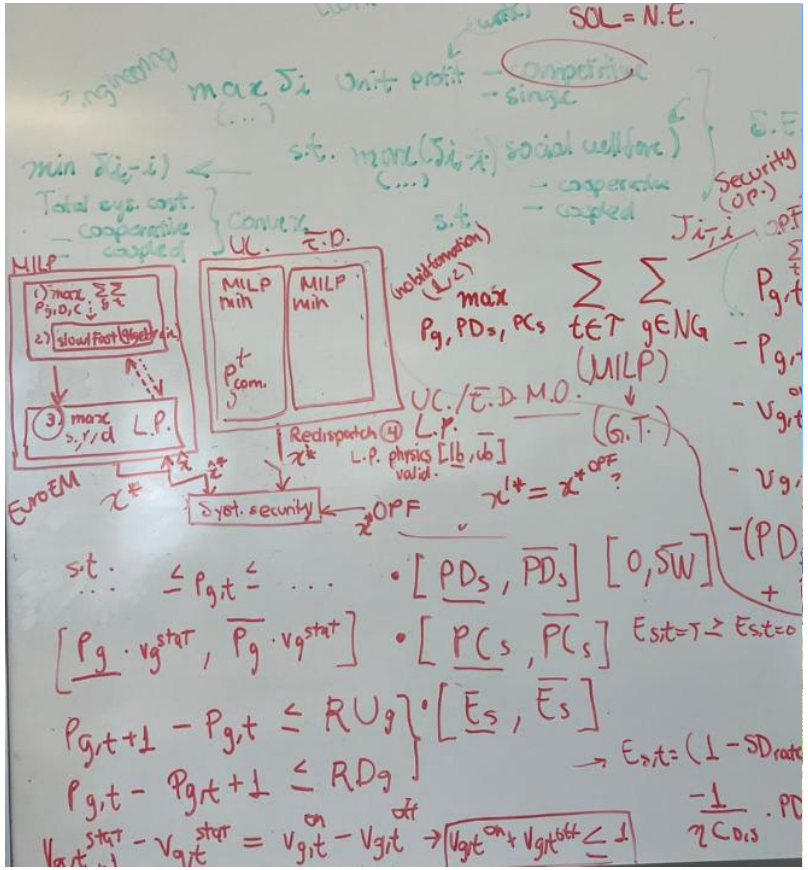
$$0 \leq SW_{s,t} \leq SW_s^{max}, \quad \forall\{s,t\} \tag{1k}$$

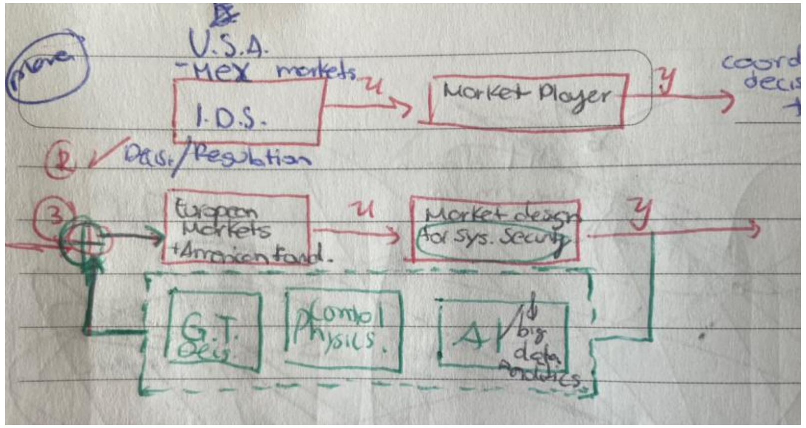
$$E_{s,t=T} \geq E_{s,t=0}, \quad \forall s \tag{1l}$$

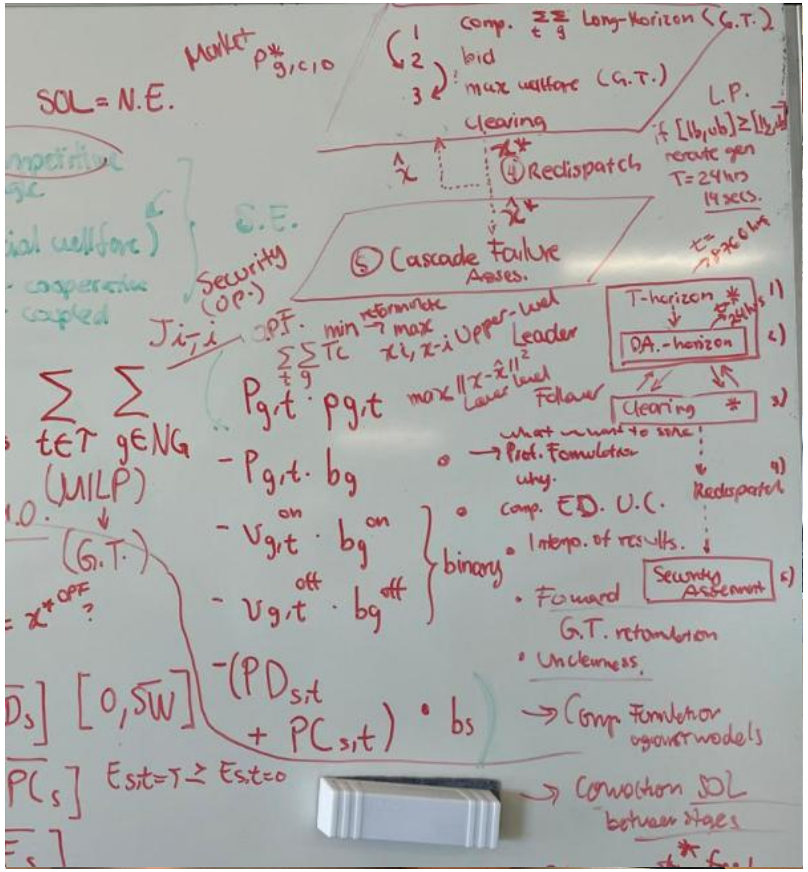
The storage unit constraints are presented with Eqs. 1g - 1l, where Eqs. 1g - 1h describe the technical operating limits. The minimum and maximum discharge/charge limits are presented

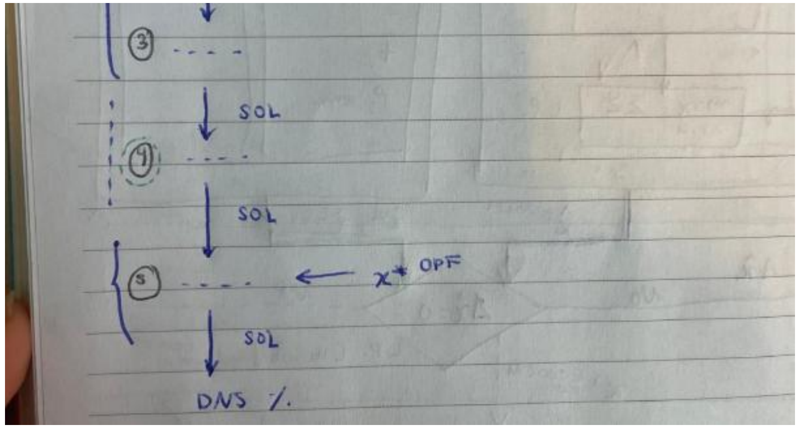


These notes document the reasoning process behind a critical review presented at ETH Zürich. Before evaluating or implementing the paper, reconstruction of its architecture was performed independently: who makes each decision, which information is available at each stage, how the optimization layers interact, and where physical and market constraints enter the model. This process allows me to test assumptions, identify missing couplings, and distinguish the paper’s essential contribution from implementation detail.









0. Abstraction and Research Process (raw cognition):

Identifying actors, decisions, information flows, optimization layers, physical constraints, assumptions, and interfaces before implementation.

Review: *A day-ahead market model for power systems: benchmarking and security implications*

Prepared for the Reliability & Risk Engineering Lab, ETH Zürich, Switzerland | 08.06.2026
Prof. Giovanni Sansavini, Dr. Blazhe Gjorgiev & RRE Team.

ETH zürich
RRE
Reliability & Risk Engineering

Stephanie Matta Bobadilla

MSc DREAM – Dynamics and Control of Renewables-Based Power Systems
KTH | UPC Barcelona | École Centrale de Nantes | UNSTPB Romania

stephanie.mattab@gmail.com

Stephanie Matta Bobadilla

Systems-level engineer

Experience across renewable energy systems, optimization, and decentralized decision-making for modern electricity networks, with exposure to both research and the electrical energy industry.

Expertise

- Power Systems
- Optimization & Game Theory
- Control Systems
- AI & Data Analytics

Current

Researcher, Decisions & Control Systems (DCS)
KTH Royal Institute of Technology, Sweden
Prof. Giuseppe Belgioioso's group.

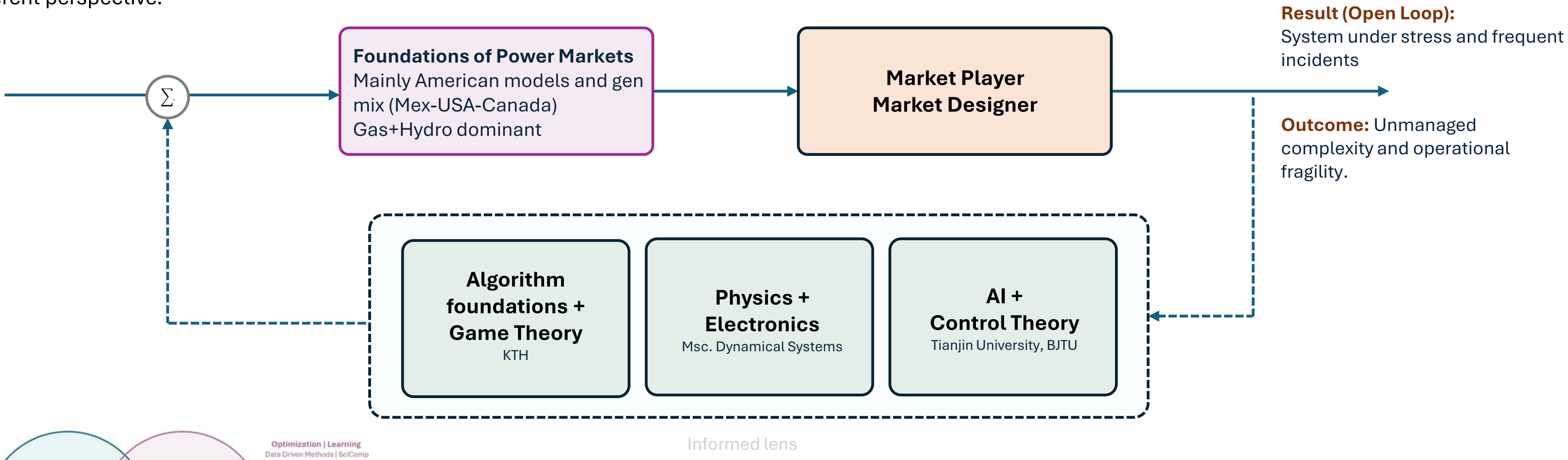
- Foundations of Multi-Agent Decision Making
- Optimization Algorithms with Real-Time Guarantees
- Control-Theoretic Perspectives on Games and Optimization

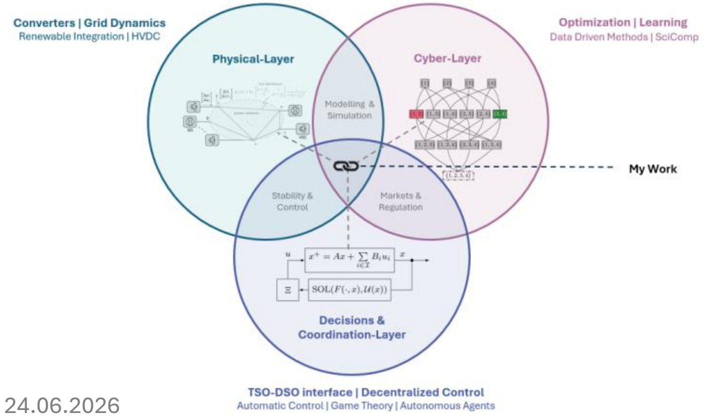
Distributed Game-Theoretic Coordination of Energy Communities under Shared Grid Constraints: Solving multiparametric Variational Inequalities for Fair Resource Allocation - Algorithms

Market and Physics

Since 2020, I have been exposed to the electricity sector primarily through markets, regulation, and energy-transition policies. This perspective allowed me to observe how decisions made at the market layer can propagate throughout the entire power system.

Over time, I complemented this view with control theory, power electronics, optimization, and multi-agent systems, gradually returning to the same questions from a different perspective.





Review

A day-ahead market model for power systems: benchmarking and security implications [1]

Andrej Stankovski, Blazhe Gjorgiev, James Ciyu Qin, and Giovanni Sansavini, Member, IEEE, 2026.
Reliability & Risk Engineering Laboratory
ETH Zürich, Switzerland

Outline

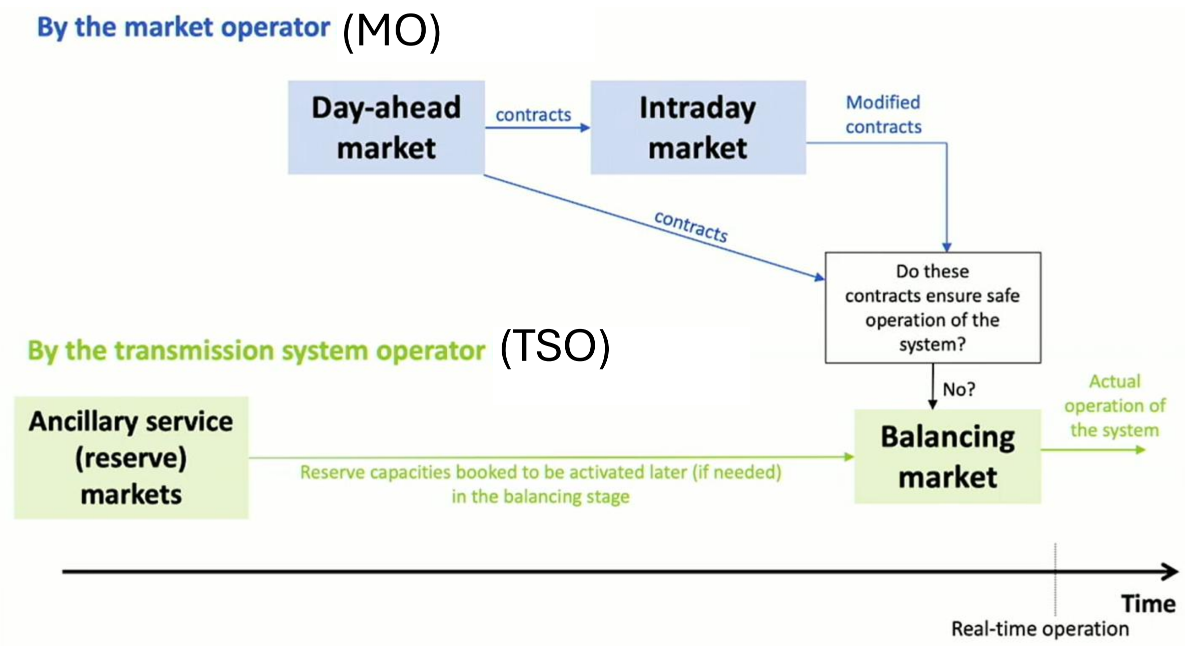
1. Wholesale Electricity Markets
2. Problem Formulation
3. Evidence
4. Method Structure
5. Model Description
6. Solutions and Interactions
7. Results
8. Review & Research strengthening
9. Conclusions

[1] Stankovski, A., et al. (2026). A day-ahead market model for power systems: benchmarking and security implication. RRE ETH Zurich. Preprint Arxiv: https://arxiv.org/pdf/2602.11842

Wholesale Electricity Markets

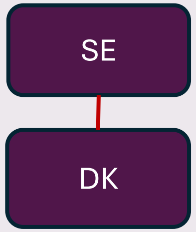
Prof. Kazempour, J. (2025). Lecture 2: Fundamentals of Electricity Markets, DTU, Denmark. Available Online at: https://www.youtube.com/watch?v=wtWW26wUDHQ

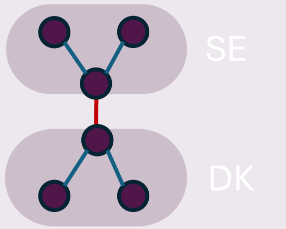
European Markets



Comparison with American markets

EU		USA
2 different entities: TSO	MO	1 single integrated: ISO
Clears reserve	Clears DAM & Intraday	Clears both simultaneously
e.g. Sweden and Denmark Bidding Zones representation 2 bidding zones		Nodal representation: 6 nodes
Internalized technical constraints into bids Thermal dynamic (ramps), block orders		Units submit all technical constraints to ISO, ISO clears then





Problem Formulation

How does the choice of market model affect security conclusions?

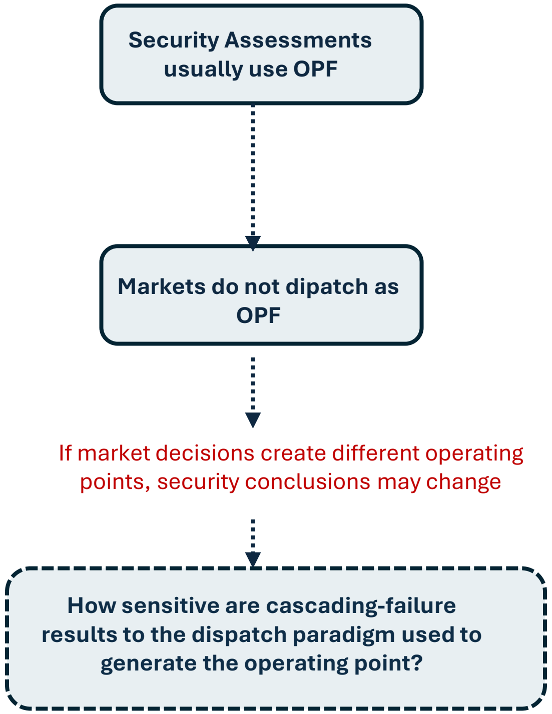
Current Practice
Security assessments commonly rely on Optimal Power Flow (OPF), Unit Commitment (UC), or Economic Dispatch (ED) generated dispatches.

The impact of market decisions on security outcomes remains unclear.

Benchmark:

- OPF
- UC
- ED
- Proposed DAM Model EuroEM

within a common security assessment framework.



Evidence

Can changing market modeling assumptions **reveal** latent system vulnerabilities *a priori* that remain hidden under traditional dispatch paradigms?

Investigation on system imbalances in Germany in June 2019 [2]

Imbalances eq.

$$[P_m - P_e]\frac{\omega_0}{2H} = \frac{d^2\delta}{dt^2}$$

Kundur, P. (1994)

Report from November 19th, 2019
(analysis from August 2019)

"German Transmission System Operators (TSOs) were forced to resolve a system imbalance of nearly **6,000 MW** caused by a sharp increase in intraday prices following an inaccurate renewable forecast.
TSOs fully activated the available reserves, shed demand, and procured balancing energy from neighboring countries.

Only six days later, a similar incident occurred, causing an imbalance of nearly **10,000 MW**."

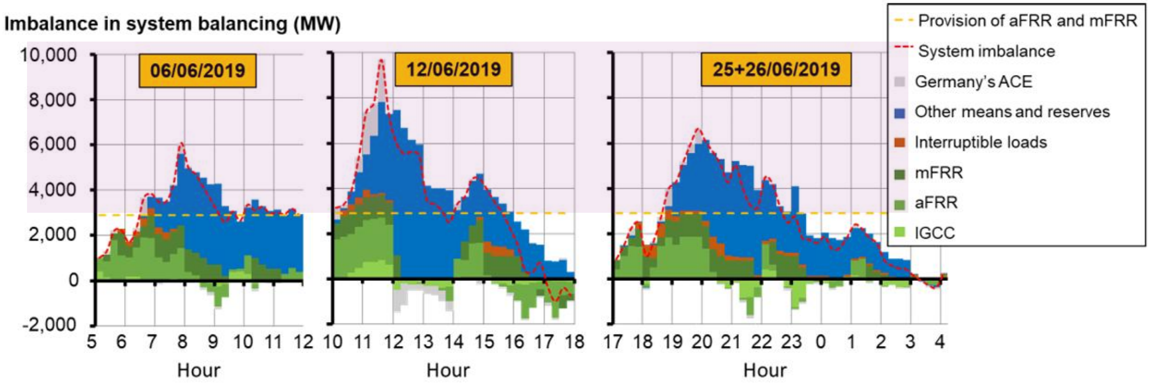


Fig. 1. Time course of the system imbalance and balancing measures activated on 06/06/2019, 12/06/2019 and 25/06/2019 (source: analysis by German TSOs)

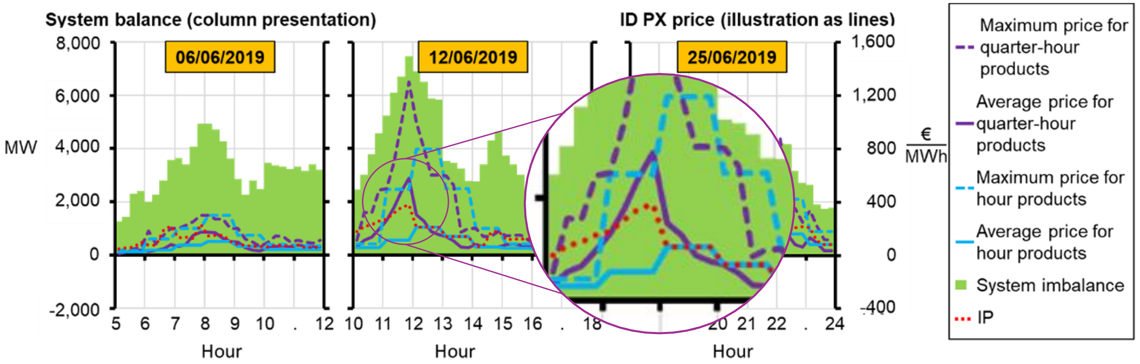


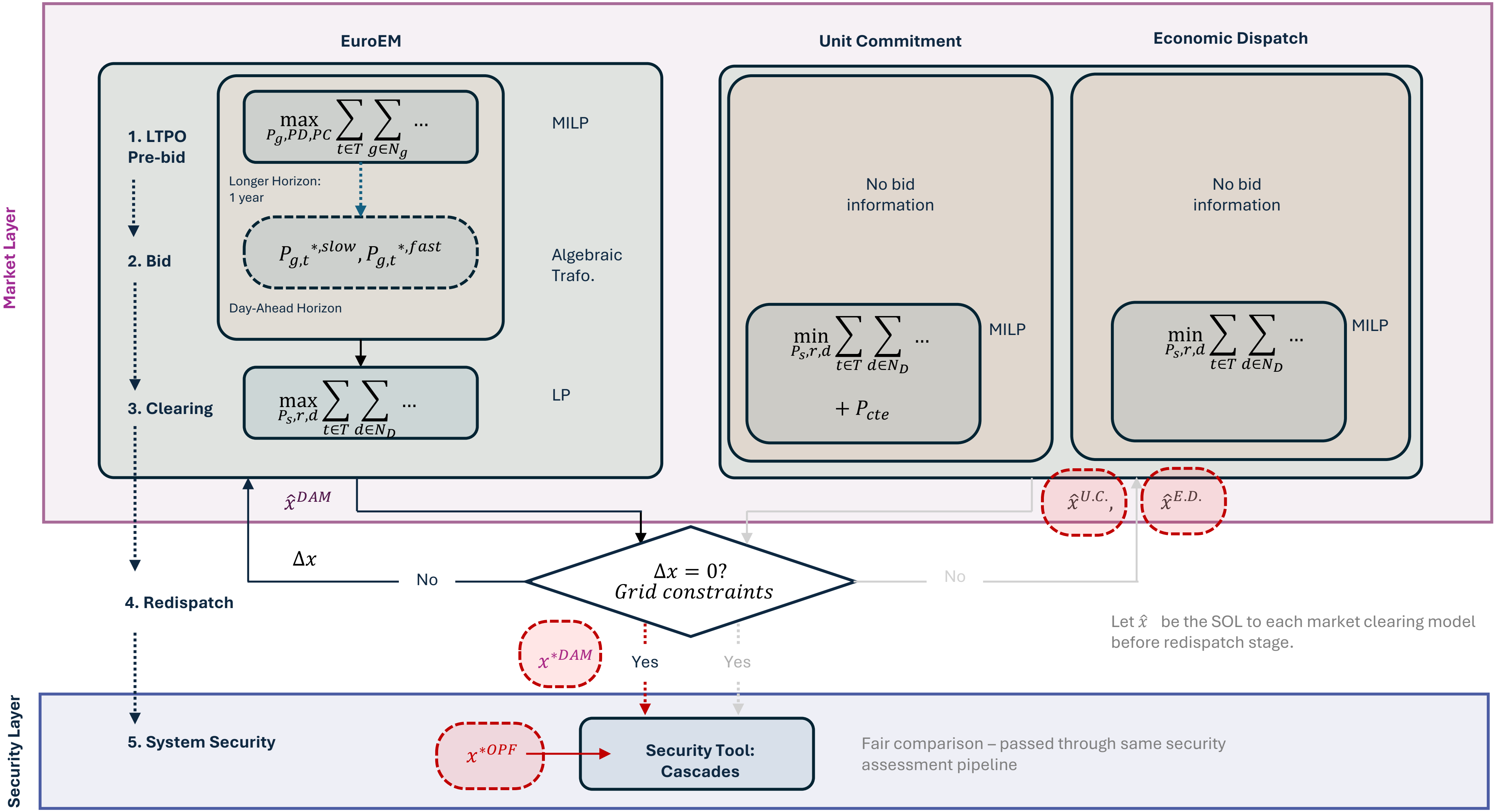
Fig. 2. Correlation between imbalance and ID exchange price (source: German TSOs)

Imbalance Price (IP) not following closely critical operation.

- 6000 MW imbalance
- 10000 MW imbalance
- Reserves exhausted
- Emergency imports
- Load Shedding

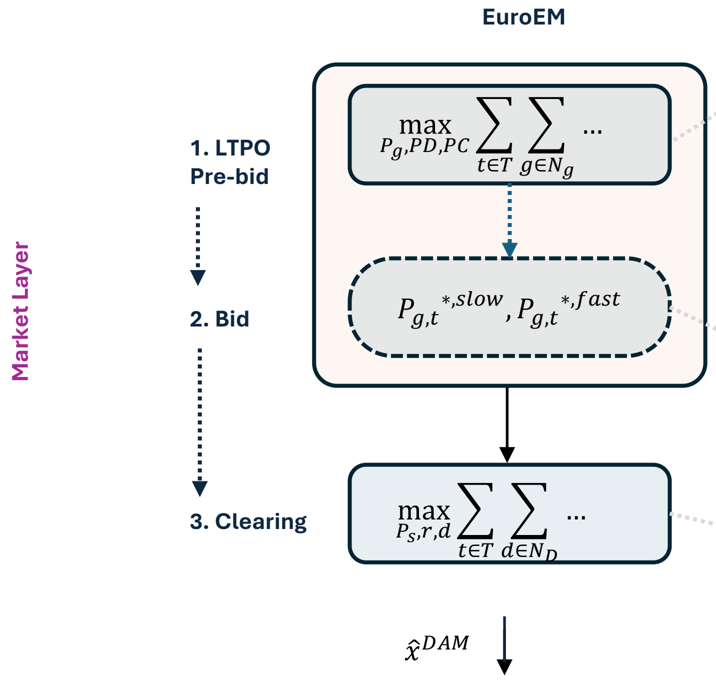
[2] Amprion. (2019). Investigation on system imbalances in Germany in June 2019. https://allemagne-energies.com/wp-content/uploads/2019/07/study-balancing-state-june-2019.pdf

Method Structure: Intuition of decision structure: 2 layers (1: market, 2: security), layer 1 comparing DAM against 2 conventional clearing methods (UC, ED), redispatch algorithm connects layer 1-2.



Market Model Description (1-3)

How and what exactly are we modelling?



1. LTPO Pre-bid

What each unit maximizes (8760 hours)
Units solves independetly, against its own Price forecast.

Pg,t · pg,t REVENUE	**Generation × own price forecast**
− Pg,t · bg MARGINAL COST	**Variable cost per MWh generated** bg = VOM + dynamic fuel cost + CO_2 emission cost
− v^{on}g,t · b^{on}g STARTUP COST	**Binary: 1 only at the hour unit turns on** v^{on}g,t ∈ {0,1} — penalizes cold starts
− v^{off}g,t · b^{off}g SHUTDOWN COST	**Binary: 1 only at the hour unit shuts down** v^{off}g,t ∈ {0,1} — with Eq.1f: v^{on} + v^{off} ≤ 1

Storage units (Added to thermal sum)

(PDs,t − PCs,t) · ps,t ARBITRAGE REVENUE	**Net discharge × price forecast** Discharge earns revenue; charging costs revenue. Storage optimizes buy-low / sell-high cycles over the year.
− (PDs,t + PCs,t) · bs VOM COST	**Wear cost on every MW moved (in or out)** bs = variable O&M Discourages unnecessary cycling without binary variables

2. Bid (ramps)

3. Clearing

What the market maximizes (24 hours)
Run by market operator
Units sumbit bids, market max (consumer surplus – producer surplus)

Dd,t · bd CONSUMER SURPLUS	**Demand cleared × willingness to pay** bd = demand bid price — maximizing this serves consumers Dd,t ≤ D^{sch},t (cannot clear more than scheduled demand)
PCs,t · σ^{Pc} STORAGE CHARGING VALUE	**Charging priced at demand-shedding cost** σ^{Pc} = bds — pushes storage to front of merit order Storage acts as price-taker: always charged when price is low
− Pg,t · bg GENERATION COST	**Producer cost subtracted from welfare** Same bg as LTPO — marginal cost sets merit order position Negative bids cleared at −bg (unit pays to produce at P^{min})
− PDs,t · σ^{PD} DISCHARGE COST	**Storage discharge sell bids priced at 0 €/MWh** σ^{PD} = 0 — storage always willing to discharge at any price Cleared when market has space; profit realized via price spread

Profit-driven bid

J1 Maximize: unit profit (1+2)

$\max_{P_g, PD, PC} \sum_{t \in T} \sum_{g \in N_g}$ sell energy at expected price – production cost – start/stop costs
maximize arbitrage revenue (buy low, sell high) while accounting for cycling costs , and pay cycling/VOM cost

J2 Maximize: social welfare (3)

$\max_{P_s, r, d} \sum_{t \in T} \sum_{d \in N_D}$ serve demand with high willingness-to-pay
- production costs to maximize total social welfare

Multi-level Solutions and Interaction (1→2 →3 →4 →5 →Result)

Information propagation through the market-physics-security pipeline

Information

1. LTPO Pre-bid

Inputs: Price forecasts, technical limits, costs

Solves: MILP max expected operating profit

Output: Prefered schedules

Intentions

2. Bid

Input: Prefered schedules

Computes: bid quantities

Output: bid limits (bounds)

Bids

3. Clearing

Inputs: bid limits, renewables offers, crossborder capacities

Solves: LP max social welfare

Output: accepted quantities $\hat{x}^{DAM}$ AND market clearing prices (dual variables)

Market allocation

4. Redispatch

Input: DAM dispatch

Solves: least-cost redispatch, s.t. DC network constraints

Output: feasible operating point, x^{*DAM}

Feasible physics

5. System Security

Input: feasible operating point x^{*DAM}

Solves: AC Power Flow, Contingencies, Cascade propagation

Output: Risk metrics i.e. Demand Not Served (DNS), Branch failures, risk curves

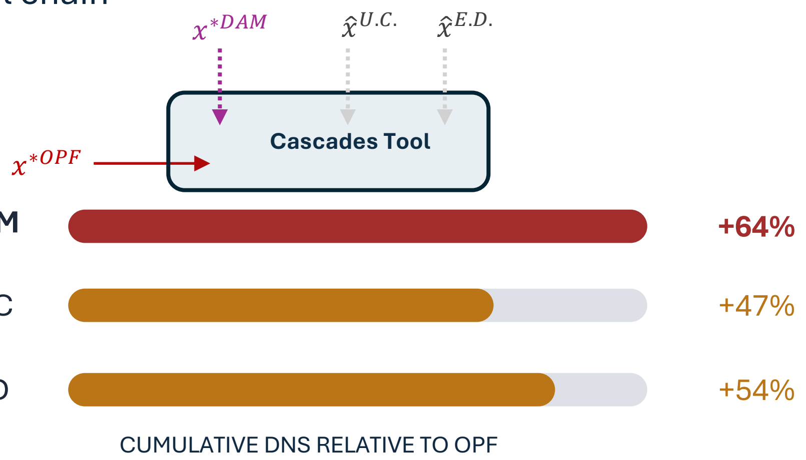
Risk

How close is x^{*DAM} to failure?

How resilient is x^{*DAM} to contingencies?

Results and Interpretation

Causal chain



Result 1: Does EuroEM actually changes from ED/UC?
Answer: Yes. Behavioral.

Observations

- Gas: activated much more frequently, becomes marginal unit more often
- Storage: charges much more, exploits arbitrage
- Hard coal: less forced generation

Av. Price:
51.2 €/MWh → 51.2 €/MWh
(+2.7%)

Hrs above 100€/MWh:
42 → 182
(4.3× increase)

Average prices barely move, tail moves.

Result 2: Does the new market create a different operating point?
Answer: Yes. Storage → more domestic flexibility → less cross-border utilization → 15% lower redispatch cost

Observations

- The market looks better, cheaper redispatch, more flexibility.

counterintuitively

Result 3: Security changes.
Redispatch costs decreases → Security decreases with it.

Observations

For DAM vs. OPF

- Adjacent branches ↑
- Intra-zonal branches ↑
- DNS ↑ 64%

Result 4: Risk changes.
Small disturbances ≈ roughly similar.
Large disturbances → very different.

Observations

- DAM changes the risk distribution.

Critical Review

The paper highlights that the dispatch paradigm used to initialize security studies can fundamentally change the estimated cascading risk, even when average market indicators change only modestly.

* Market participants optimize against exogenous price forecasts rather than endogenous market prices.

Strengths

- **Methodological contribution**
  Integrates market-informed dispatch with cascading failure analysis through a common security assessment framework.

- **Realistic market representation**
  Captures decentralized, profit-driven bidding before centralized market clearing, providing a more realistic benchmark than OPF-based dispatch assumptions.

- **Controlled experimental design**
  Uses the same redispatch and cascading assessment across all dispatch paradigms, isolating the impact of the market model on security.

Modeling Assumptions

Behavioral

- Participants optimize independently (price takers).
- No endogenous strategic interaction or market power

Market scope

- The framework is limited to the day-ahead market. Intraday, balancing and ancillary service markets are outside the scope.

Uncertainty

- Price uncertainty represented through noisy forecasts.
- No stochastic or robust optimization.

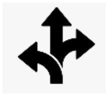
Security scope

- The security assessment focuses on thermal overload-induced cascading failures.
- Frequency dynamics, voltage instability and protection system complexity are outside the scope.

Research oportunities

Strategic market behavior
Endogenize bidding decisions using equilibrium or bilevel formulations.

Multi-market coordination
Include intraday, balancing and ancillary service markets.

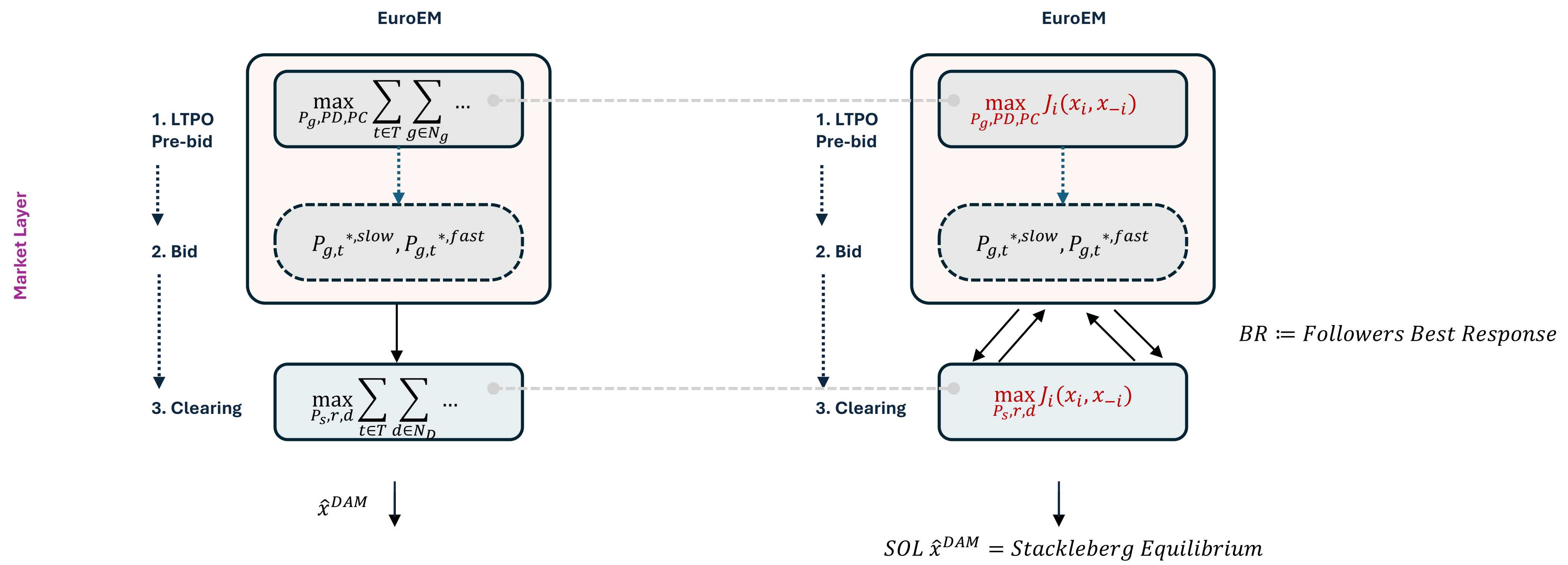


Integrated market-security co-design
Investigate market designs that jointly optimize economic efficiency and system resilience.

Possible extension: Endogenous strategic behavior:
Competition and Bilevel Stackelberg formulation

Competition: e.g. Cournot, SOL = Nash Equilibria (NE)

*Can sequential vs bilevel-fashion reveal new insights on **strategic interactions**?*
How can strategic interaction threat system security?



* Expensive bilevel formulations can be accelerated by several computational *tricks*, i.e. 2-phase algorithms with warm start: Offline equilibria pre-computation, Online piecewise-affine (PWA) control law decision-making, as one of the examples. Available development exploration in: https://github.com/StephanieMattaB/DyNECT

Conclusions

Overall, the paper provides a valuable benchmark for market-informed security assessment while opening opportunities to incorporate richer market behavior and uncertainty representations

From observations:

Behavioral market models change dispatch → Dispatch changes operating points → Operating points change estimated security

Economic

- Gas ↑
- Storage ↑
- Price spikes ↑
- Average prices ≈

The market behaves differently.

Operation

- Domestic flexibility ↑
- Redispatch ↓
- Operating point changes

The grid operates differently.

Security

- Adjacent branch failures ↑
- Intra-zonal failures ↑
- DNS ↑ 64%
- Risk curve shifts right

The same security assessment produces a different risk profile.

What could we have infered upfront? More risk revealed.

New insights from study?

Average prices almost unchanged –> Behaviour changed
OPF worst for tiny events (more branch failures) -> better for catastrophic events.

What is it telling us?

New AND different spatial distribution of energy & risk.
Market modeling assumptions are themselves part of the security assessment, not merely a preprocessing step.

Thank you for your attention and invitation!

Stephanie Matta Bobadilla
stephanie.mattab@gmail.com

Dr. Blazhe Gjorgiev, Prof. Giovanni Sansavini & RRE Team.

24.06.2026